PROJECT TITLE

QUANTUM ERROR MITIGATION FOR NOISY QUANTUM COMPUTING SYSTEMS

Use Case Title

Improving the Reliability of Quantum Circuits Using Digital Zero-Noise Extrapolation

Real-Life Scenario

A quantum application is executed on a near-term quantum processor to solve a task such as optimization, classification, or simulation. The circuit produces a result, but hardware noise causes the measured output to differ from the ideal answer. Instead of waiting for large-scale fault-tolerant quantum computers, a software-level mitigation layer can be placed between the quantum program and the final result. Digital Zero-Noise Extrapolation (ZNE) provides such a use case: the same logical computation is executed with different effective noise levels, and the observations are used to estimate the result that would have been obtained at zero noise. This makes error mitigation relevant to practical NISQ-era quantum applications.

Problem Statement

Current NISQ hardware is affected by gate errors, decoherence, calibration uncertainty and other noise sources. These effects accumulate as circuits become deeper, reducing the reliability of expectation values and measurement probabilities. Full quantum error correction requires substantial hardware resources and is not yet practical for many near-term applications. Therefore, there is a need for techniques that can improve quantum results using the hardware and gate-level access already available. The project investigates how digital ZNE can reduce the error of noisy computations and how the choice of scaling and extrapolation method affects the final result.

How Quantum Computing Applies Here

- Digital Zero-Noise Extrapolation: The circuit is evaluated at several effective noise levels and the measured values are extrapolated toward the zero-noise limit.
- Unitary Folding: A gate or circuit can be combined with its inverse and the original operation. Ideally the logical action is unchanged, while the additional physical operations increase exposure to noise.
- Noise Scaling: Different scaling factors are used to generate multiple observations. Digital scaling can operate at the instruction or gate level without requiring detailed pulse-control access.
- Extrapolation: Linear, polynomial, Richardson, exponential or other models can be fitted to the measured data to estimate the value at the zero-noise point.
- Hybrid Workflow: The classical computer controls circuit generation, fitting and analysis, while the quantum processor or simulator performs the noisy circuit evaluations.

Everyday / Practical Impact of This Use Case

- Quantum applications can obtain more reliable expectation values without waiting for full fault-tolerant hardware.
- Researchers can compare mitigated and unmitigated results to understand how strongly a circuit is affected by noise.
- The method can be incorporated into software workflows using gate-level access, making it suitable for experiments on near-term platforms.
- The same logical algorithm can be tested under several effective noise levels, giving a practical way to study the relationship between noise and output quality.
- The approach can be used with randomized circuits as well as variational optimization problems, allowing the method to be evaluated across different circuit types.

Current Limitations / Challenges

- Noise scaling increases the number of circuit operations or executions, so mitigation introduces additional computational and measurement overhead.
- Extrapolation is an estimation problem; statistical fluctuations in measured data can make high-order models unstable.
- Unitary folding is not equally suitable for every error source. For example, errors associated with state preparation and measurement are not naturally scaled by increasing circuit depth.
- Coherent or systematic errors may not be amplified in the same way as incoherent noise, which can limit the effectiveness of folding-based scaling.
- The best extrapolation method depends on the circuit, noise model, scaling factors and available measurement budget; no single method is guaranteed to dominate in every situation.
- Present-day NISQ devices have limited qubit counts and non-negligible noise, so results from small demonstrations should not automatically be interpreted as large-scale fault-tolerant performance.

Evaluation Plan

The proposed evaluation compares a noisy circuit with its mitigated version. A suitable reference result can be obtained from an ideal or sufficiently accurate simulation. For each circuit, results are collected at multiple noise-scale factors. The project can compare linear, polynomial and exponential extrapolation, while also examining Richardson extrapolation where appropriate. Performance can be measured using absolute error, error-reduction ratio, expectation-value deviation, circuit depth, gate count and the number of circuit executions. This makes the study a comparison of both accuracy and resource cost.

Conclusion

This use case demonstrates how quantum error mitigation can be applied to a realistic NISQ-era quantum workflow. Digital ZNE does not remove the physical noise itself; instead, it uses controlled noise scaling and classical inference to estimate a less noisy result. The supplied research paper shows that unitary folding and different extrapolation strategies can produce improvements in suitable experiments, while also highlighting statistical and hardware-related limitations. The proposed project will therefore focus on measuring the practical benefit of ZNE and identifying the trade-off between error reduction and additional resources.

Literature Survey

The following research literature provides the foundation for the proposed digital zero-noise extrapolation use case. The first paper is the research paper supplied for this project; the remaining entries summarize closely related quantum error-mitigation directions.

Paper	Problem Statement	Quantum Approach / Algorithm	Key Findings / Results	Relevance to Our Use Case
Paper 1 – Giurgica-Tiron et al. (2021) Digital zero noise extrapolation for quantum error mitigation	NISQ processors produce noisy results, while scalable fault-tolerant error correction is not yet broadly available.	Digital ZNE, unitary folding, parameter noise scaling, polynomial, linear, Richardson, exponential and adaptive extrapolation.	The study reports strong error reduction in benchmark experiments and demonstrates ZNE on randomized circuits and a QAOA/MAXCUT example. It also shows that extrapolation performance varies by method.	Direct foundation for our project: it supplies the ZNE workflow, scaling ideas and comparison methods to be evaluated.
Paper 2 – Temme et al. (2017) Error mitigation for short-depth quantum circuits	Short quantum circuits can still accumulate errors that affect observable estimates.	Probabilistic error cancellation / quasi-probability based mitigation.	Shows that noisy quantum computations can be improved through classical post-processing, while the method has sampling overhead.	Provides a comparison point for ZNE: both mitigate errors without requiring full fault-tolerant correction, but use different principles.
Paper 3 – Li et al. (2017) Efficient variational quantum simulator incorporating active error minimization	Variational quantum algorithms are sensitive to hardware errors and can give degraded optimization results.	Error-mitigation ideas applied to variational quantum computation.	Demonstrates that error-aware processing can improve the usefulness of variational computations on noisy hardware.	Supports testing ZNE with variational circuits rather than only simple benchmark circuits.
Paper 4 – Cai et al. (2023) Quantum error mitigation	Different QEM methods have different assumptions, resource costs and applicability to noise sources.	Review of ZNE, probabilistic error cancellation, symmetry-based methods, virtual distillation and related QEM techniques.	Provides a broad framework for understanding the capabilities and limitations of quantum error mitigation.	Provides the wider literature foundation for positioning our ZNE comparison among modern QEM techniques.

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Primary research source supplied: Tudor Giurgica-Tiron, Yousef Hindy, Ryan LaRose, Andrea Mari and William J. Zeng, "Digital zero noise extrapolation for quantum error mitigation," arXiv:2005.10921.